

MLII ECG Anomaly Detection and Arrhythmia Classification

Final one-lead temporal-holdout system

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The study makes two decisions from one ECG lead

Decision sequence

- 1 Detect whether the current MLII beat is normal or abnormal.
- 2 If abnormal, classify PVC, PAC, LBBB, RBBB, or AFib.

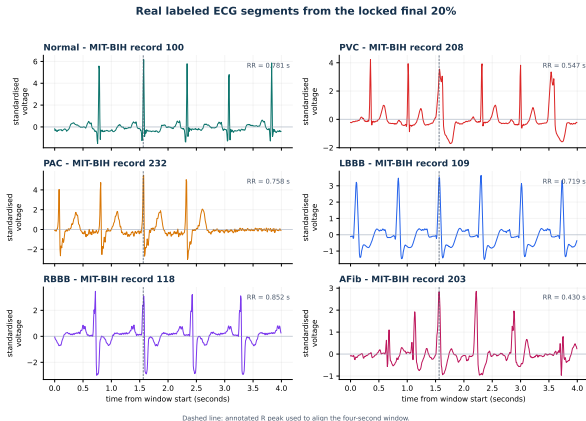
Claim boundary

The result measures later ECG from known MITDB patients. It is not an unseen-patient or external-hospital validation.

The target was more than 90% whole-system accuracy, with a subject-clustered 95% confidence interval whose lower bound also exceeds 90%.

Source: [1, 2]

The problem is not a simple ECG-threshold task

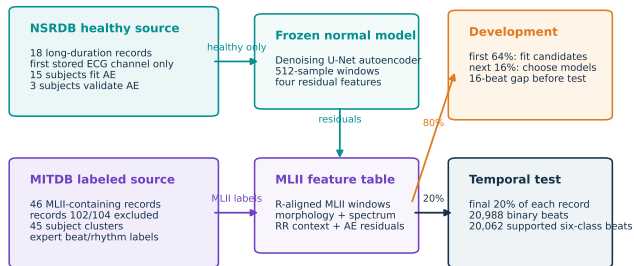


PVC, LBBB, and RBBB mainly alter morphology; PAC and AFib also need rhythm context. The model therefore combines MLII shape, RR timing, and normal-reconstruction mismatch.

Source: [3]

Normal-reference and arrhythmia data have separate roles

Dataset roles and evaluation boundaries



The temporal test uses later signal from the same MITDB patients; it is not an unseen-patient test.

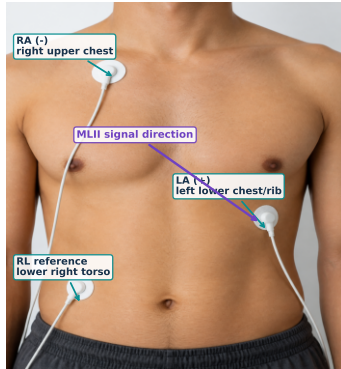
NSRDB creates the frozen normal autoencoder. MITDB MLII creates the supervised feature table and the final temporal test.

Source: [4, 3]

The supervised system is strictly MLII-only

Lead positions and one-channel rule used by the model

A. Realistic MLII torso electrode placement



B. What each model actually receives

NORMAL AUTOENCODER
18 NSRDB subjects
first stored channel only
WFDB name: ECG1
placement not documented

ABNORMALITY DETECTOR
MITDB MLII only
MLII selected by signal name
record 114: MLII is channel 2
records 102/104 excluded

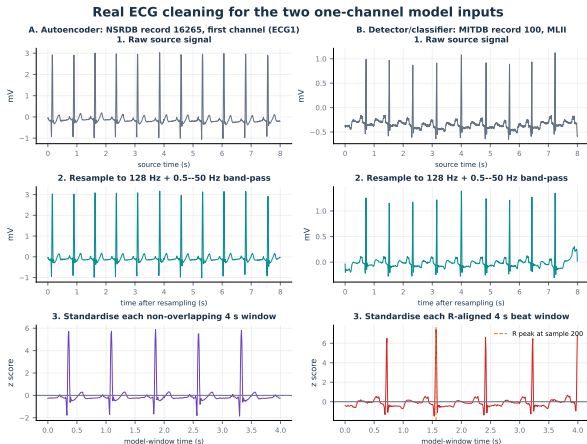
SUBTYPE CLASSIFIER
same MLII window
same 242-value feature vector
same four autoencoder residuals
no second ECG lead is used

Therefore the performance numbers describe MLII temporal continuation, not a two-lead or unseen-patient result.

The autoencoder uses the first stored NSRDB ECG channel. The detector, subtype classifier, temporal test, and installed replay use MLII selected by signal name.

Source: [4, 3]

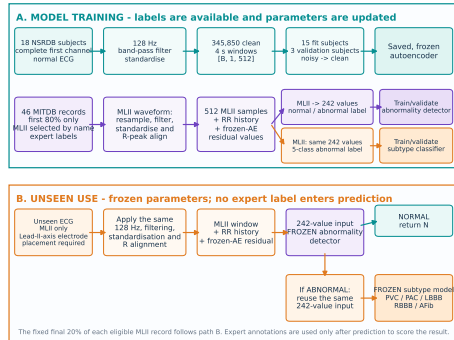
Cleaning keeps the two signal roles explicit



Both pipelines output one 512-sample input, but the healthy NSRDB reference channel and the MITDB MLI1 deployment channel are not treated as the same lead.

Training and use follow one fixed preparation route

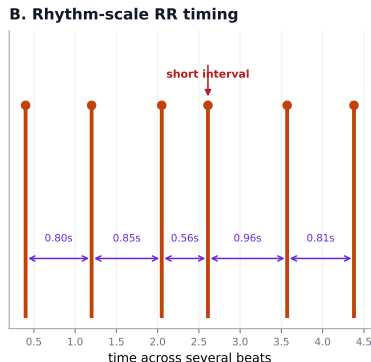
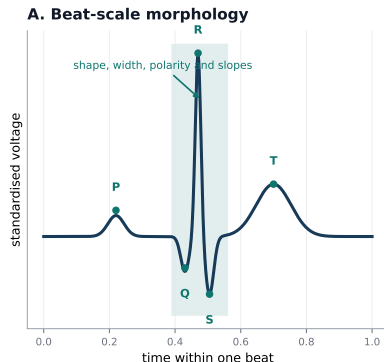
Data preparation during training and during unseen use



The same resampling, filtering, standardisation, R alignment, RR calculation, and feature order are used during development and installed use.

The model uses morphology and rhythm evidence together

Two ECG evidence scales used by the final model



The final feature vector joins these scales with four normal-reconstruction residuals.

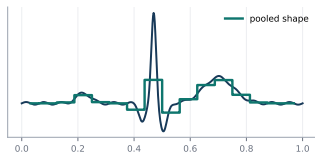
Morphology describes the local beat shape; RR timing describes where that beat occurs in the rhythm sequence.

Source: [1, 5]

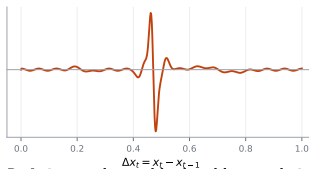
Raw ECG becomes compact numerical evidence

Transparent operations that create the final feature vector

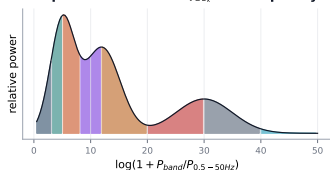
A. Pooling keeps shape and reduces small shifts



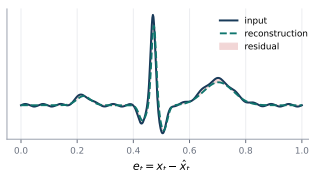
B. Derivatives describe QRS sharpness



C. Spectral bands summarise frequency content



D. Autoencoder residuals add normal-structure evidence



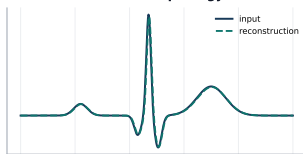
Pooling, derivatives, frequency power, and statistics preserve useful MLII evidence while reducing sensitivity to small shifts and noise.

Source: [6, 7]

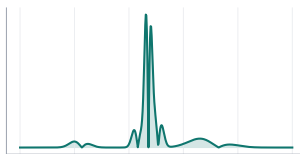
Normal reconstruction is evidence, not the diagnosis

Normal-only reconstruction: residual is evidence, not diagnosis

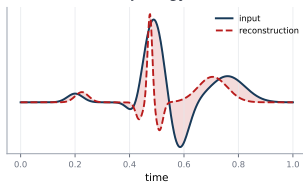
A. Familiar normal morphology



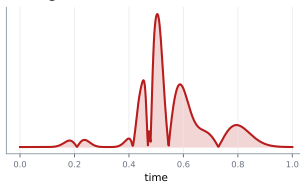
B. Small residual



C. Unfamiliar morphology



D. Larger residual feature



The supervised detector receives four residual summaries and decides together with morphology and RR evidence.

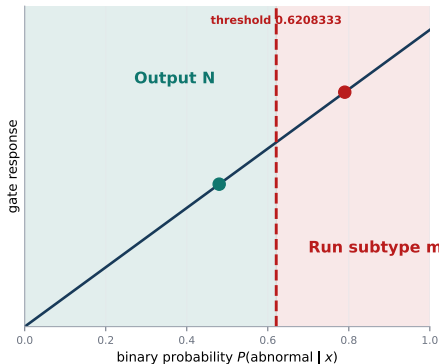
A larger residual may indicate novelty, but the final decision is made by supervised tree models that also see MLII morphology and RR timing.

Source: [8, 9, 10]

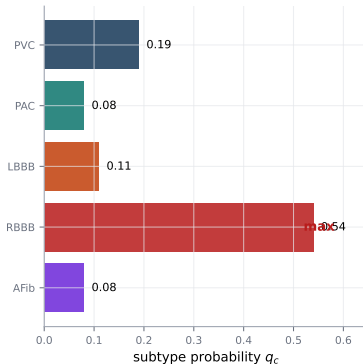
The hierarchy separates abnormality from subtype

Hierarchical probability logic in the final model

A. Binary abnormal gate



B. Conditional subtype selection

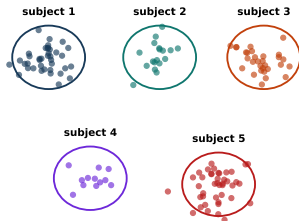


The subtype model is conditional: it is used only when the binary gate passes the abnormality threshold.

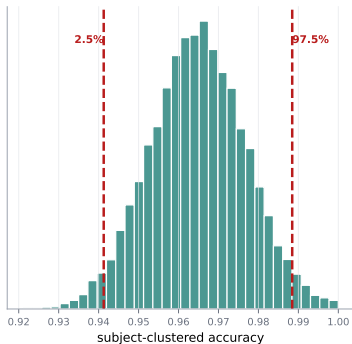
Validation must respect patient clusters

Why validation must respect subject clusters

A. Many beats, but fewer independent subjects



B. Bootstrap resamples subjects, not individual beats



This supports a known-patient temporal-continuation claim, not an unseen-patient claim.

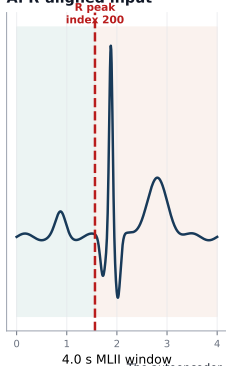
Subject-clustered resampling prevents many beats from one record being treated as many independent people.

Source: [11]

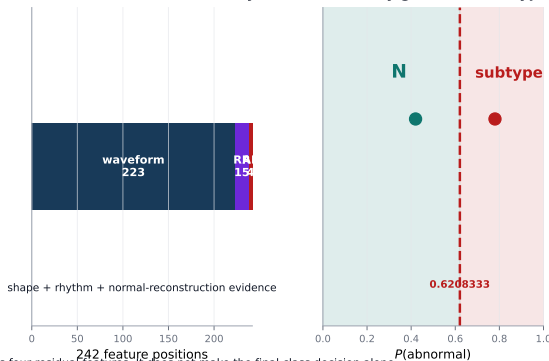
The final runtime system is a one-lead hierarchy

Final ECG system: from one MLII beat to one decision

A. R-aligned input



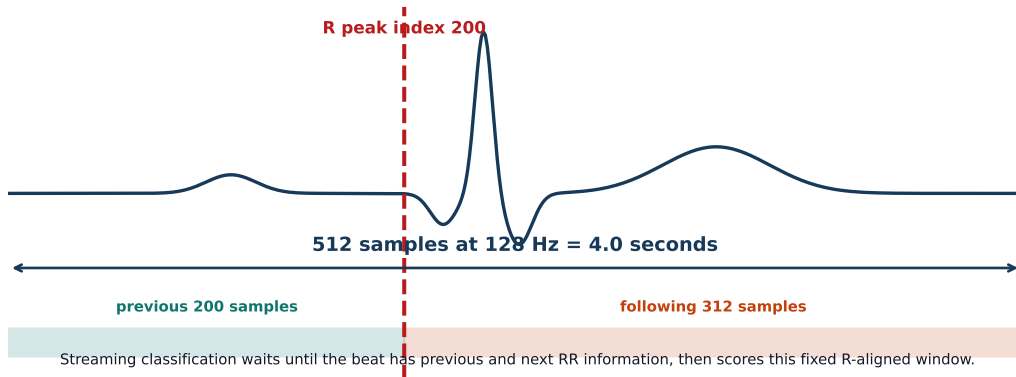
B. Feature vector reused by both ensemble gate then subtype



The autoencoder supplies four residual features. It does not make the final class decision alone.

For each detected MLII heartbeat, the saved system builds one 242-value vector, applies the abnormality gate, and conditionally applies the five-class subtype model.

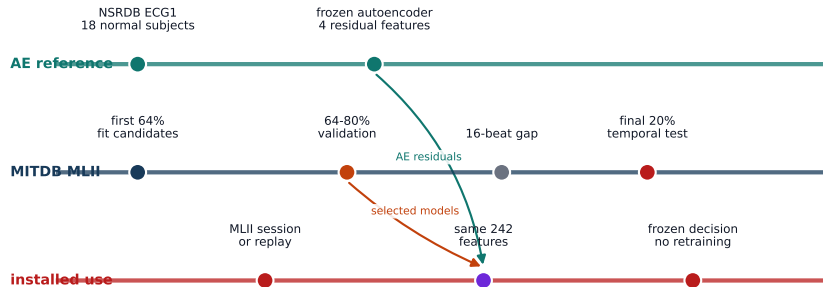
Each beat is scored from an R-aligned window



The R peak is fixed at sample index 200 inside a 512-sample, four-second window, giving both pre-QRS and post-QRS context.

Training, testing, and installed use are separated

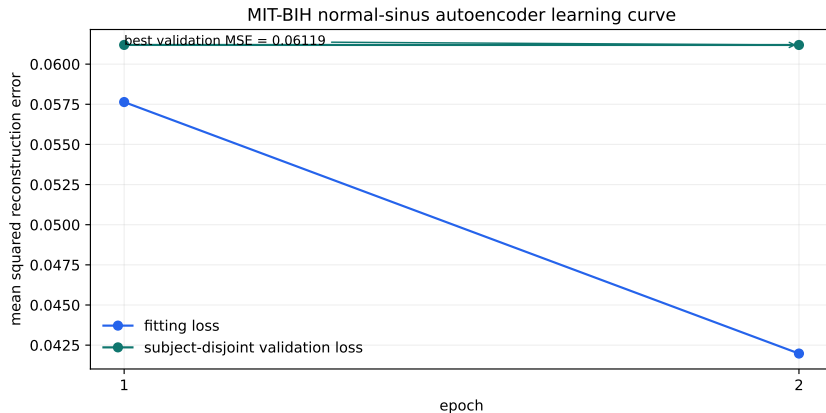
Training, testing, and installed use of the final bundle



The performance claim is known-patient temporal continuation: later MLII from patients whose earlier MLII was used for development.

Live/session use applies the frozen bundle. It does not update the autoencoder, the detector, or the subtype classifier.

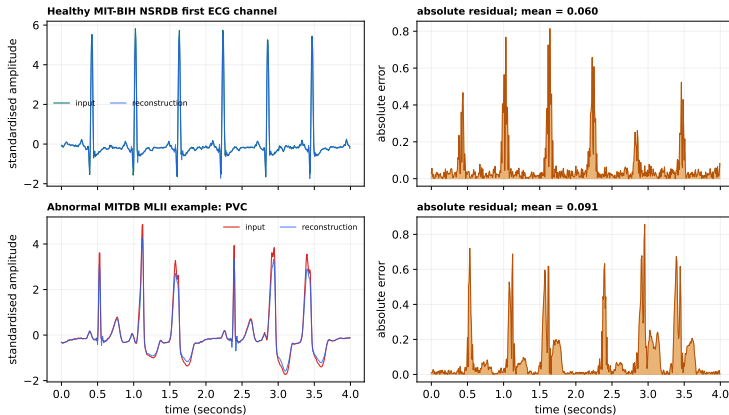
The autoencoder checkpoint is frozen before supervised fitting



The normal-only autoencoder used 288,673 fitting windows from 15 NSRDB subjects and 57,177 validation windows from three different subjects.

Autoencoder mismatch becomes four residual features

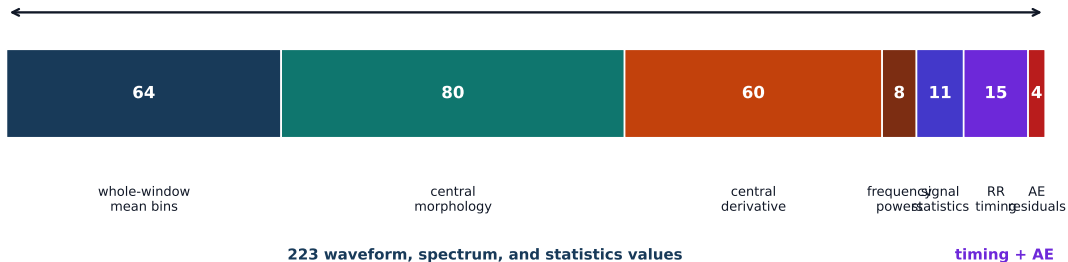
Real one-channel examples through the frozen normal autoencoder



Whole-window MAE, whole-window MSE, central morphology MAE, and residual-change MAE are appended to the supervised feature vector.

Both supervised stages receive the same 242 values

242 values used by the binary detector and subtype classifier

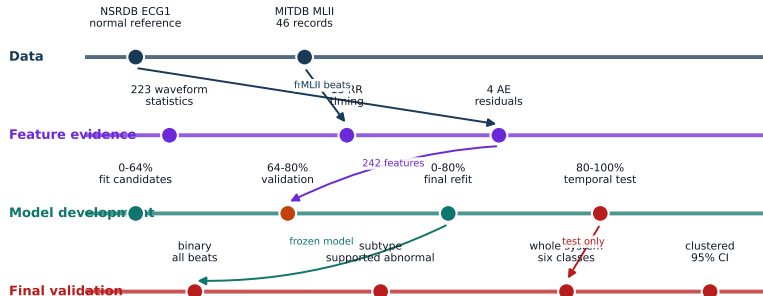


223 MLII waveform/statistical values, 15 RR timing values, and 4 autoencoder residual values form the final evidence vector.

The final experiment keeps selection away from the test

Experimental method for the final temporal-holdout run

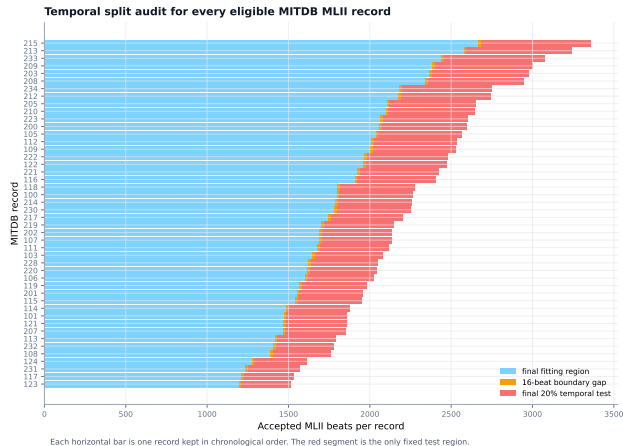
One-lead MLII experiment: select with earlier beats, freeze decisions, score only the final 20% timeline.



The test timeline is not used for candidate choice, threshold choice, or final fitting in this rerun.

Candidate models are selected on earlier temporal regions, refit before the test boundary, and scored only on the final 20% timeline.

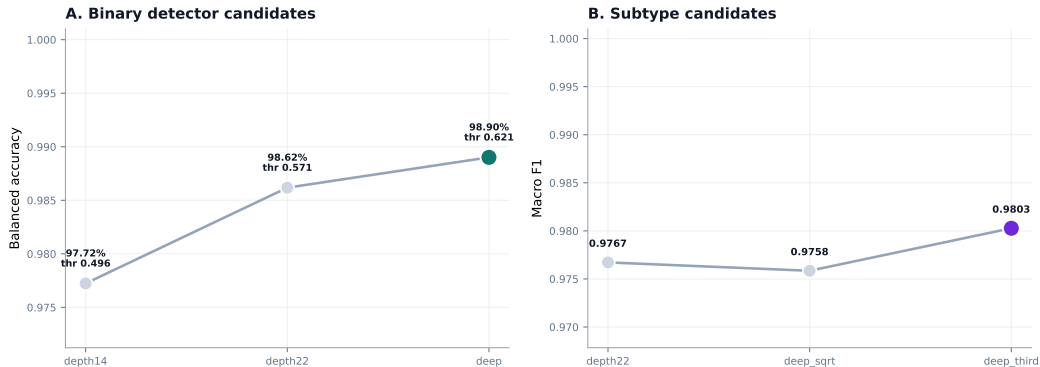
Every eligible record follows the same temporal split



Blue beats are available for final fitting, yellow beats are the 16-beat boundary gap, and red beats are the held-out final temporal test.

Validation selected the final Extra Trees pair

Validation-region model selection



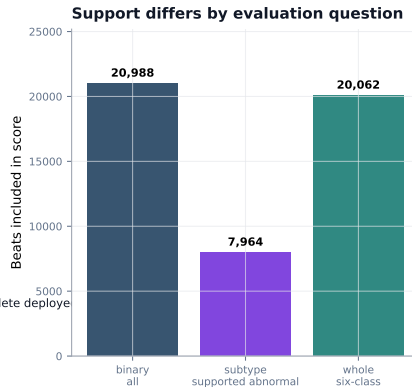
The selected detector uses 240 trees and threshold 0.6208333. The selected subtype model uses 260 trees with one-third feature sampling.

Three evaluations answer different questions

Evaluation order and test support



Separate supports identify where errors happen: abnormal gate, conditional subtype name, or complete deployment



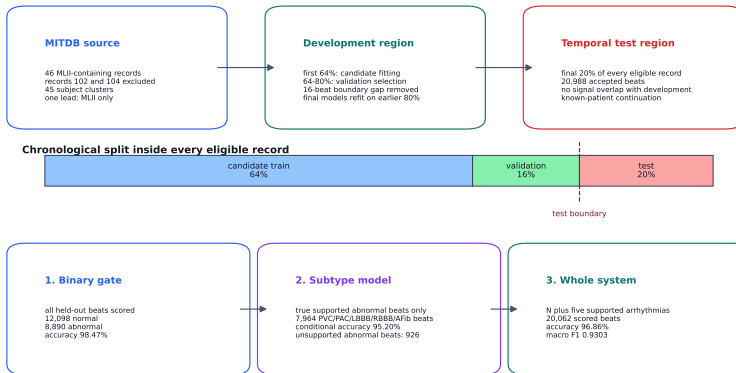
Binary detection uses all held-out accepted beats. Conditional subtype scoring uses true supported abnormal beats.

Whole-system scoring uses N plus the five supported abnormal classes.

The fixed test is later ECG from known patients

How the final system was tested

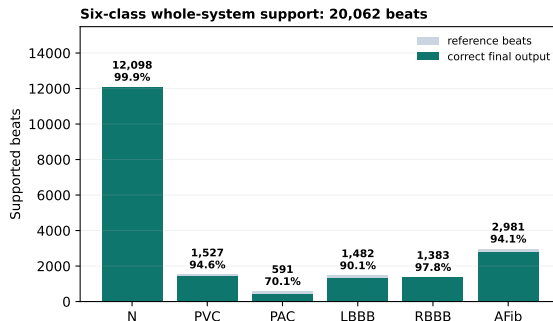
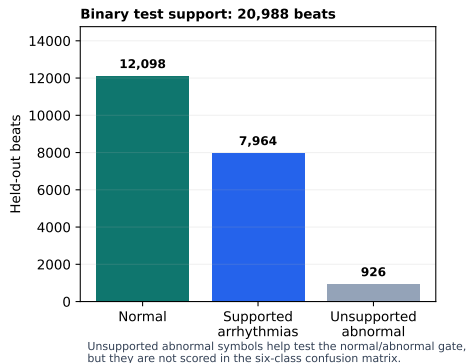
Fixed within-patient temporal holdout: earlier MLII from each patient trains/selects the model; only the final 20% timeline is scored.



The final 20% of each eligible MITDB MLII record is held out for scoring; earlier ECG from the same records remains available for development.

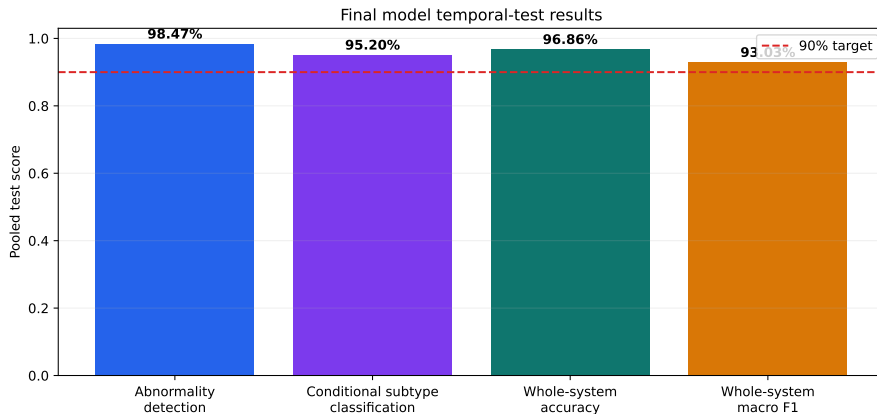
The whole-system test contains six supported outputs

What data was used for the final temporal test



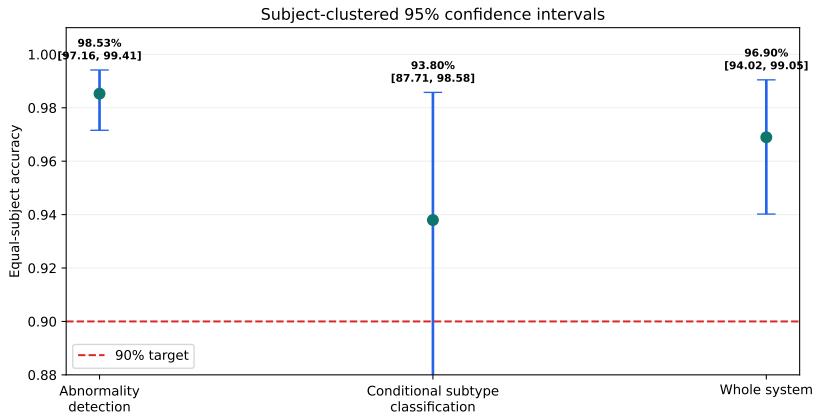
The six-class whole-system test contains 20,062 supported beats: N, PVC, PAC, LBBB, RBBB, and AFib. Unsupported abnormal beats remain in binary testing only.

Temporal whole-system accuracy is 96.86%



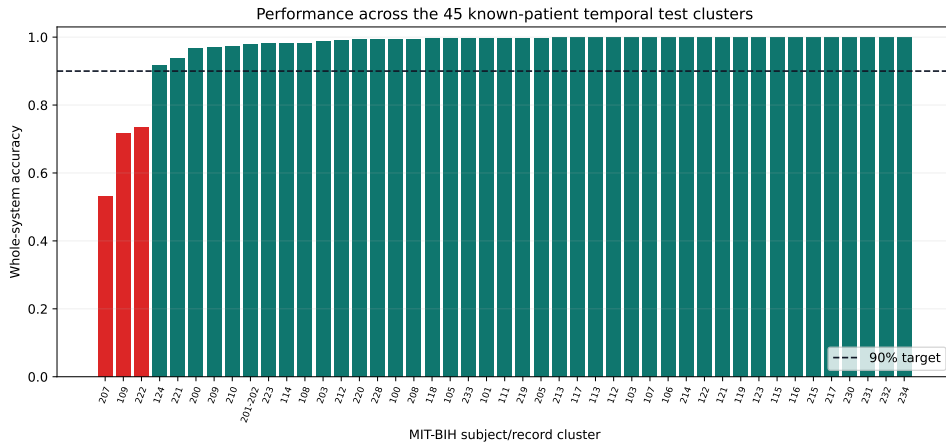
Binary accuracy is 98.47%, conditional subtype accuracy is 95.20%, whole-system accuracy is 96.86%, and whole-system macro F1 is 0.9303.

The 95% interval clears 90%, but is still wide



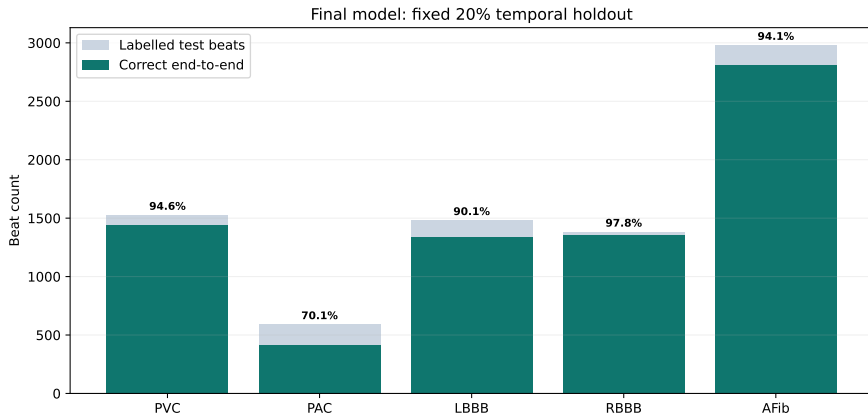
Whole-system subject-clustered 95% CI is 94.02–99.05%. The 5.03-point width does not meet the planned three-point precision target.

Most subject clusters exceed the target



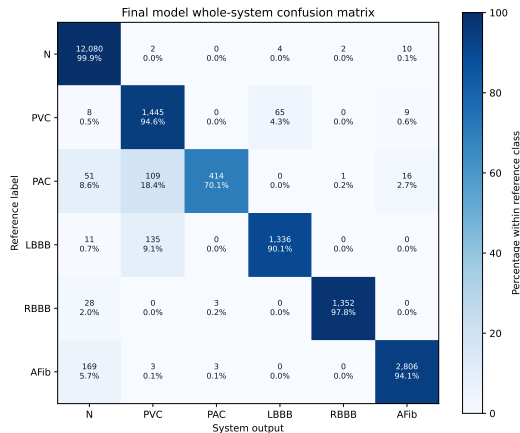
Forty-two of 45 subject clusters are above 90%; equal-subject whole-system accuracy is 96.90%.

PAC is the main unsupported recall goal



End-to-end recall is PVC 94.63%, PAC 70.05%, LBBB 90.15%, RBBB 97.76%, and AFib 94.13%.

The confusion matrix shows the remaining failure modes



The largest subtype confusions are LBBB to PVC, PAC to PVC, and PVC to LBBB. The binary gate also sends some AFib and PAC beats to normal.

The goals were reached only for the stated temporal scope

Reached Pooled whole-system accuracy exceeds 90%.

Reached The subject-clustered lower 95% bound exceeds 90%.

Not reached Confidence-interval width is 5.03 rather than at most three points.

Not reached PAC recall is 70.05%, below 90%.

Not tested Completely unseen patients or an external hospital database.

Safety conclusion

This is a research prototype, not a clinical diagnostic device.

The next study must add independent patients

- ① Keep the MLII input rule fixed across training and deployment.
- ② Add independent PAC, LBBB, RBBB, and AFib patients.
- ③ Reserve an external patient-level database for one final test.
- ④ Pre-register the lower CI bound, interval width, and per-class recall.
- ⑤ Compare the full model with and without autoencoder residual features.
- ⑥ Test real AD8232 recordings without changing the frozen model.

Source: [1]

Final conclusion

Core finding

A single ECG lead is sufficient for 96.86% whole-system accuracy on unseen later signal from known patients, with a subject-clustered 95% interval of 94.02–99.05%.

Required interpretation

The result does not prove 96.86% accuracy for a completely new patient. PAC recall and interval width remain important weaknesses.

The final software, experiment artifact, report, and presentation now describe the same one-channel architecture, with MLII fixed for supervised use.

References I

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- [2] G. B. Moody and R. G. Mark, "The impact of the mit–bih arrhythmia database," *IEEE Engineering in Medicine and Biology Magazine*, vol. 20, no. 3, pp. 45–50, 2001.
- [3] G. B. Moody and R. G. Mark, "Mit–bih arrhythmia database." PhysioNet, 2005.
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- [5] S. Mousavi and F. Afghah, "Inter- and intra-patient ecg heartbeat classification for arrhythmia detection: A sequence to sequence deep learning approach," in *IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 1308–1312, 2019.
- [6] P. Kligfield *et al.*, "Recommendations for the standardization and interpretation of the electrocardiogram: Part i," *Circulation*, vol. 115, no. 10, pp. 1306–1324, 2007.

References II

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- [8] G. E. Hinton and R. R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," *Science*, vol. 313, no. 5786, pp. 504–507, 2006.
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- [10] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Computing Surveys*, vol. 41, no. 3, pp. 1–58, 2009.
- [11] C. M. Rutter, "Bootstrap estimation of diagnostic accuracy with patient-clustered data," *Academic Radiology*, vol. 7, no. 6, pp. 413–419, 2000.